0.1 Compactness Variants: Total Boundedness, Sequential Compactness, and the Ascoli-ArzelTheorem

In the standard topology of R^n, the Heine-Borel theorem establishes that a set is compact if and only if it is closed and bounded. However, in arbitrary metric spaces, mere boundedness is insufficient to guarantee compactness. We must introduce a stricter geometric constraint.

A metric space (X, d) is **totally bounded** if for every $\epsilon > 0$, there exists a finite set of points $\{x_1, x_2, \ldots, x_n\} \subset X$ (an ϵ-net) such that $X = \bigcup_{i=1}^{n} B_\epsilon(x_i)$.

A metric space (X, d) is **sequentially compact** if every sequence $(x_n)_{n=1}^{\infty}$ in X possesses a convergent subsequence whose limit lies in X.

If a metric space (X, d) is sequentially compact, then it is totally bounded.

We proceed by contradiction. Assume (X, d) is sequentially compact but not totally bounded. Then there exists some $\epsilon_0 > 0$ for which no finite ϵ_0-net can cover X.

We construct a sequence (x_n) recursively. Choose $x_1 \in X$ arbitrarily. Because the single ball $B_{\epsilon_0}(x_1)$ cannot cover X, there exists $x_2 \in X \setminus B_{\epsilon_0}(x_1)$. This implies $d(x_1, x_2) \geq \epsilon_0$. Inductively, suppose we have chosen points $\{x_1, \ldots, x_k\}$ such that $d(x_i, x_j) \geq \epsilon_0$ for all $i \neq j$. Because no finite union of ϵ_0-balls covers X, the set $X \setminus \bigcup_{i=1}^{k} B_{\epsilon_0}(x_i)$ is non-empty. We select x_{k+1} from this complement, ensuring $d(x_{k+1}, x_i) \geq \epsilon_0$ for all $1 \leq i \leq k$.

This process yields an infinite sequence $(x_n)_{n=1}^{\infty}$ satisfying $d(x_n, x_m) \geq \epsilon_0$ for all $n \neq m$. Consequently, no subsequence can be a Cauchy sequence, as the distance between any two distinct terms is bounded below by ϵ_0. Since every convergent sequence must be Cauchy, (x_n) admits no convergent subsequence. This directly contradicts the premise that X is sequentially compact. Therefore, X must be totally bounded.

A central application of these metric space topologies lies in the analysis of function spaces. Let K be a compact metric space, and let $C(K)$ denote the Banach space of continuous real-valued functions on K, equipped with the supremum norm $\|f\|_\infty = \sup_{x \in K} |f(x)|$. To determine when a sequence of functions in $C(K)$ admits a uniformly convergent subsequence, we require the notions of uniform boundedness and equicontinuity.

A family of functions $\mathcal{F} \subset C(K)$ is **uniformly bounded** if there exists $M > 0$ such that $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and all $x \in K$. The family $\mathcal{F}$ is **equicontinuous** if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in K$ with $d(x, y) < \delta$, we have $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$.

[Ascoli-ArzelTheorem - Sufficiency] Let K be a compact metric space. If a sequence of functions $(f_n)_{n=1}^{\infty}$ in $C(K)$ is uniformly bounded and equicontinuous, then it contains a uniformly convergent subsequence.

Because K is compact, it is totally bounded and thus separable; it possesses a countable dense subset $D = \{x_1, x_2, \ldots\}$.

We employ Cantor's diagonal argument. Because the sequence is uniformly bounded, the sequence of real numbers $(f_n(x_1))$ is bounded. By the Bolzano-Weierstrass theorem, it has a convergent subsequence, which we index as $(f_{1,k})_{k=1}^{\infty}$. Next, the sequence $(f_{1,k}(x_2))$ is a bounded real sequence, yielding a convergent subsequence $(f_{2,k})_{k=1}^{\infty}$. Proceeding inductively, we construct nested subsequences $(f_{m,k})_{k=1}^{\infty}$ such that the m-th subsequence converges at the point x_m.

We extract the diagonal sequence $g_k = f_{k,k}$. For any fixed m, $(g_k)_{k\geq m}$ is a subsequence of $(f_{m,k})_{k=1}^{\infty}$. Therefore, the sequence (g_k) converges pointwise at every $x \in D$.

We now prove that (g_k) is uniformly Cauchy on K. Let $\epsilon > 0$. By the equicontinuity of the original sequence, there exists $\delta > 0$ such that $d(x, y) < \delta |g_k(x) - g_k(y)| < \frac{\epsilon}{3}$ for all $k \in N$ and $x, y \in K$. Because K is compact, it is totally bounded. The open balls $\{B_\delta(x_i)\}_{x_i \in D}$ cover K, so we can extract a finite subcover corresponding to points $\{p_1, \ldots, p_r\} \subset D$.

Since (g_k) converges at each p_i, it is a Cauchy sequence at each of these finitely many points. Thus, there exists an $N \in N$ such that for all $m, k \geq N$ and all $1 \leq i \leq r$:

$$|g_k(p_i) - g_m(p_i)| < \frac{\epsilon}{3}$$

Let $x \in K$ be arbitrary. The point x must lie in $B_\delta(p_i)$ for some $1 \leq i \leq r$. For $m, k \geq N$, we apply the triangle inequality:

$$|g_k(x) - g_m(x)| \leq |g_k(x) - g_k(p_i)| + |g_k(p_i) - g_m(p_i)| + |g_m(p_i) - g_m(x)|$$

Since $d(x, p_i) < \delta$, equicontinuity strictly bounds the first and third terms by $\frac{\epsilon}{3}$. The middle term is bounded by $\frac{\epsilon}{3}$ due to the pointwise Cauchy convergence at p_i. Therefore:

$$|g_k(x) - g_m(x)| < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

Since this bound is independent of x, (g_k) is uniformly Cauchy. Because $C(K)$ is complete under the supremum norm, (g_k) converges uniformly on K.